Sampling Distribution

Def'n: A statistic is any function of a random sample

For example, $\bar{x}$, s, ... are statistics.

	Distribution/population	sample	
parameters	μ	$\bar{x}$	statistics
	σ^2	s^2	
	σ	s	

The distribution of a statistic is called a sampling distribution.

Example

For a six-faced die.

Let X = result of a roll.

The pmf of x is $f(x) = P(X=x) = \frac{1}{6}$ $\quad x = 1, \dots 6$

$$\mu = \sum_{x \in S_x} x f(x) = 1 \times \frac{1}{6} + 2 \times \frac{1}{6} + \dots + 6 \times \frac{1}{6} = \boxed{3.5}$$

$$\sigma^2 = E(X^2) - [E(X)]^2 = \sum_{x \in S_x} x^2 f(x) - (3.5)^2$$

$$= 1^2 \times \frac{1}{6} + 2^2 \times \frac{1}{6} + \dots + 6^2 \times \frac{1}{6} - (3.5)^2 = \boxed{2.9167}$$

a) Suppose roll 2 times

Let $\bar{X}$ = mean of 2 rolls $= \frac{X_1 + X_2}{2}$

Find the sampling distribution of $\bar{X}$.

$\bar{x}$	1	1.5	2	2.5	3	3.5	4	4.5	5	5.5	6
$P(\bar{X} = \bar{x})$	$\frac{1}{36}$	$\frac{2}{36}$	$\frac{3}{36}$	$\frac{4}{36}$	$\frac{5}{36}$	$\frac{6}{36}$	$\frac{5}{36}$	$\frac{4}{36}$	$\frac{3}{36}$	$\frac{2}{36}$	$\frac{1}{36}$